2024년도 2학기

소프트웨어 공학

Homework #2

Structured Analysis for the RVC Control Software

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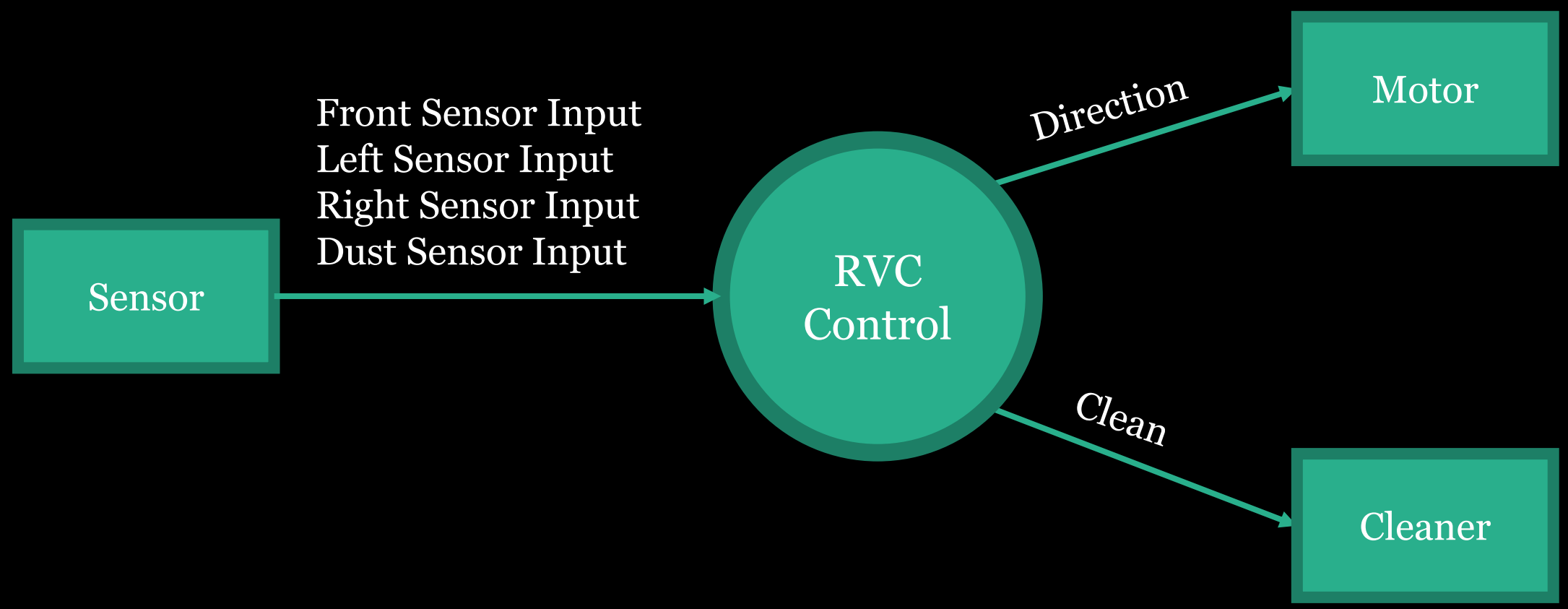
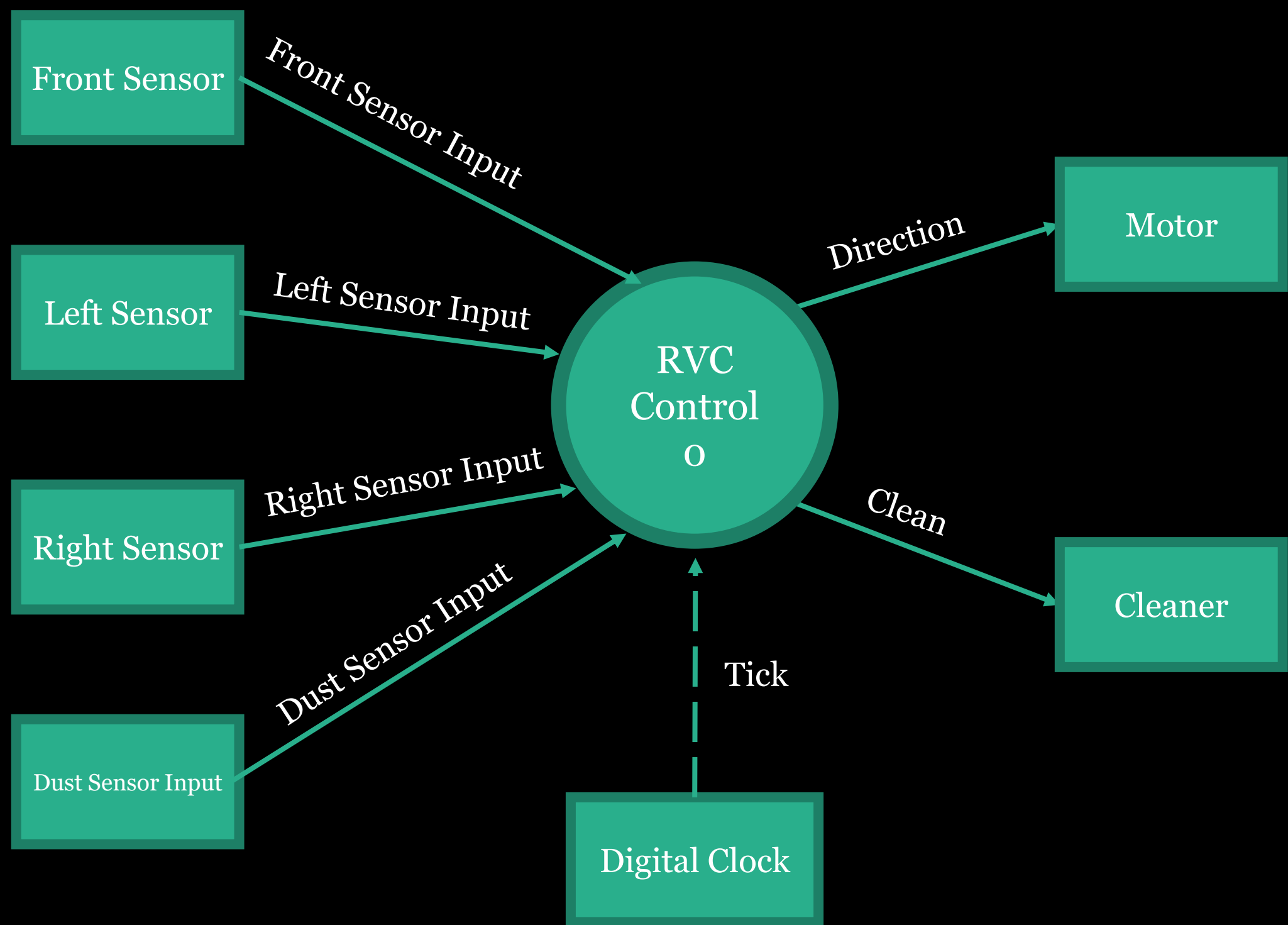
System Context Diagram



Event List

Input / Output Event	Description
Front Sensor Input	Detects obstacles in front of the RVC
Left Sensor Input	Detects obstacles in the left side of the RVC periodically
Right Sensor Input	Detects obstacles in the right side of the RVC periodically
Dust Sensor Input	Detects dust on the floor periodically
Direction	Direction commands to the motor (go forward / go backward / turn left with an angle / turn right with an angle)
Cleaner	Turn off / Turn on / Power-Up

DFD Level 0


Front Sensor
Front Sensor Input
Left Sensor
Left Sensor Input
Right Sensor
Right Sensor Input
Dust Sensor Input
Dust Sensor Input
RVC Control 0
Direction
Motor
Clean
Cleaner
Tick
Digital Clock

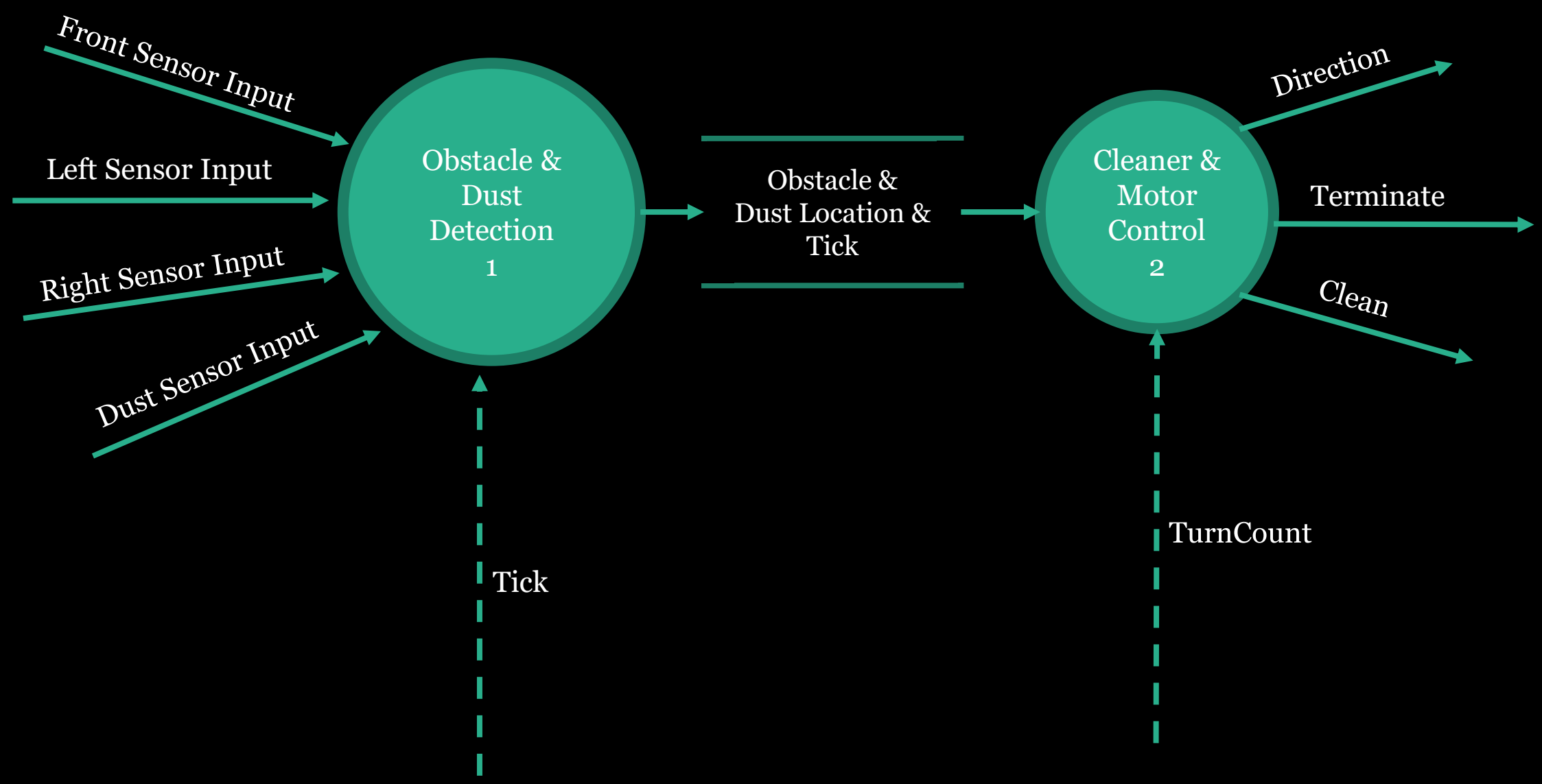
DFD Level 0 - Process Specification

Reference No.	0
Name	RVC Control
Input	Front Sensor Input, Left Sensor Input, Right Sensor Input, Dust Sensor Input, Tick
Output	Direction, Clean
Process Description	정해진 주기대로 측정된 센서의 Input 값에 따라, Direction, Clean에 Output 데이터를 전송합니다.

DFD Level 0 - Data Dictionary

Input / Output Event	Description	Format / Type
Front Sensor	Detects obstacles in front of the RVC	True/False, Interrupt
Left Sensor	Detects obstacles in the left side of the RVC periodically	True/False, Periodic
Right Sensor	Detects obstacles in the right side of the RVC periodically	True/False, Periodic
Dust Sensor	Detects dust on the floor periodically	True/False, Periodic
Direction	Direction commands to the motor (go forward / turn left with an angle / turn right with an angle / Turn backward)	Forward / Left / Right / Stop / Backward
Cleaner	Turn off / Turn on / Power-Up	On/Off/Up

Level 1


Front Sensor Input
Left Sensor Input
Right Sensor Input
Dust Sensor Input
Obstacle & Dust Detection 1
Obstacle & Dust Location & Tick
Cleaner & Motor Control 2
Direction
Terminate
Clean
Tick
TurnCount

DFD Level 1 -
Process Specification

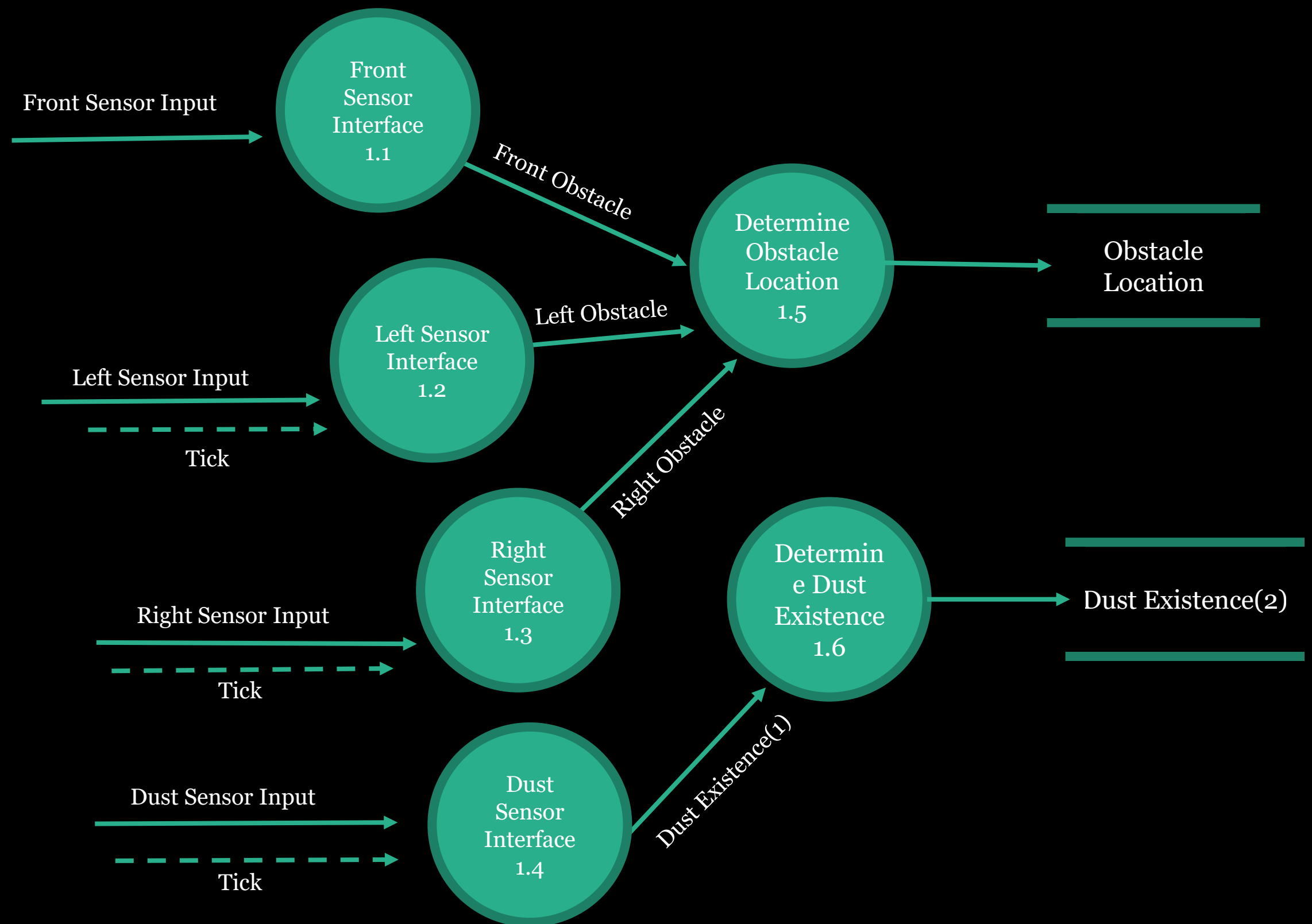
Reference No.	1
Name	Obstacle & Dust Detection
Input	Front Sensor Input, Left Sensor Input, Right Sensor Input, Dust Sensor Input, Tick
Output	Obstacle & Dust Location & Tick
Process Description	정해진 주기마다 측정된 센서의 Input 값과 Tick을 Obstacle & Dust Location & Tick에 저장

Reference No.	2
Name	Cleaner & Motor Control
Input	Obstacle & Dust Location & Tick, TurnCount
Output	Direction, Clean , Terminate
Process Description	TurnCount가 일정 수치 이상으로 올라가면 Terminate, Obstacle & DustLocation & Tick 에 저장되어있는 센서값, Tick을 계산하여 모터의 방향과 작동 여부를 결정하고, Cleaner의 터보 작동과 작동 여부를 결정.

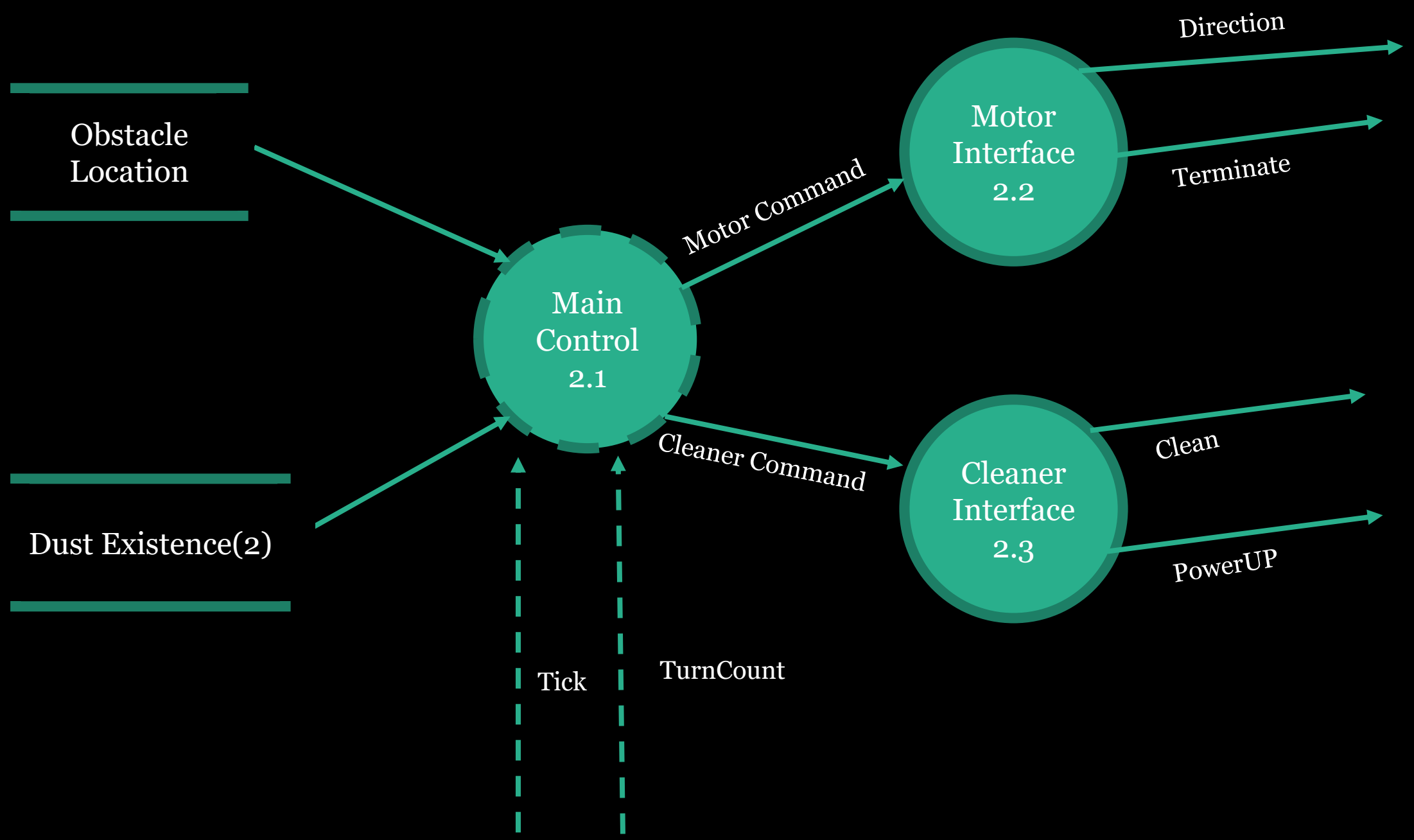
DFD Level 1 - Data Dictionary

Input / Output Event	Description	Format / Type
Front Sensor Input	Detects obstacles in front of the RVC	True/False, Interrupt
Left Sensor Input	Detects obstacles on the left side of the RVC periodically	True/False, Periodic
Right Sensor Input	Detects obstacles on the right side of the RVC periodically	True/False, Periodic
Dust Sensor Input	Detects dust on the floor periodically	True/False, Periodic
Direction	Determines whether motors work or not, and direction of motors (go forward / turn left with an angle / turn right with an angle / Turn backward)	Forward / Left / Right / Stop / Backward
Cleaner	Turn on / Turn off / Power-Up	On/Off/Up
TurnCount	Records the number of times the RVC Rotate in a row, to prevent an infinite Loop. If the RVC goes forward, initialized to zero	Integer

Level 2


Front Sensor Input
Front Sensor Interface 1.1
Front Obstacle
Left Sensor Input
Tick
Left Sensor Interface 1.2
Left Obstacle
Right Sensor Input
Tick
Right Sensor Interface 1.3
Right Obstacle
Dust Sensor Input
Tick
Dust Sensor Interface 1.4
Dust Existence(1)
Determine Obstacle Location 1.5
Obstacle Location
Determine Dust Existence 1.6
Dust Existence(2)

Level 2


Obstacle Location
Dust Existence(2)
Main Control 2.1
Motor Command
Motor Interface 2.2
Direction
Terminate
Cleaner Command
Cleaner Interface 2.3
Clean
PowerUP
Tick
TurnCount

DFD Level 2 -
Process Specification

Reference No.	1.1
Name	Front Sensor Interface
Input	Front Sensor Input, Tick
Output	Front Obstacle
Process Description	항상 작동중인 Front Sensor의 Input 값을 Front Obstacle에 저장

Reference No.	1.2
Name	Left Sensor Interface
Input	Left Sensor Input, Tick
Output	Left Obstacle
Process Description	주기적으로 측정하는 Left Sensor의 Input 값을 Left Obstacle에 저장

DFD Level 2 -
Process Specification

Reference No.	1.3
Name	Right Sensor Interface
Input	Right Sensor Input, Tick
Output	Right Obstacle
Process Description	주기적으로 측정하는 Right Sensor의 Input 값을 Right Obstacle에 저장

Reference No.	1.4
Name	Dust Sensor Interface
Input	Dust Sensor Input, Tick
Output	Dust Existence
Process Description	주기적으로 측정하는 Dust Sensor의 Input 값을 Dust Obstacle에 저장

DFD Level 2 -
Process Specification

Reference No.	1.5
Name	Determine Obstacle Location
Input	Front Obstacle, Left Obstacle, Right Obstacle
Output	Obstacle Location
Process Description	Front Obstacle, Left Obstacle, Right Obstacle 입력을 받아 Obstacle Location에 저장

Reference No.	1.6
Name	Determine Dust Existence
Input	Dust Existence(1)
Output	Dust Existence(2)
Process Description	Dust Existence(1) 입력을 받아 Dust Existence(2)에 저장

DFD Level 2 -
Process Specification

Reference No.	2.1
Name	Main Control
Input	Obstacle Location, Dust Existence(2), Tick, TurnCount
Output	Motor Command, Cleaner Command
Process Description	Obstacle Location, Dust Existence(2), Tick, TurnCount의 값을 읽어와 동작을 결정해, Motor Command, Cleaner Command를 전달

Reference No.	2.2
Name	Motor Interface
Input	Motor Command
Output	Direction , Terminate
Process Description	Motor Command를 받아, Direction과 Teminate에 할당

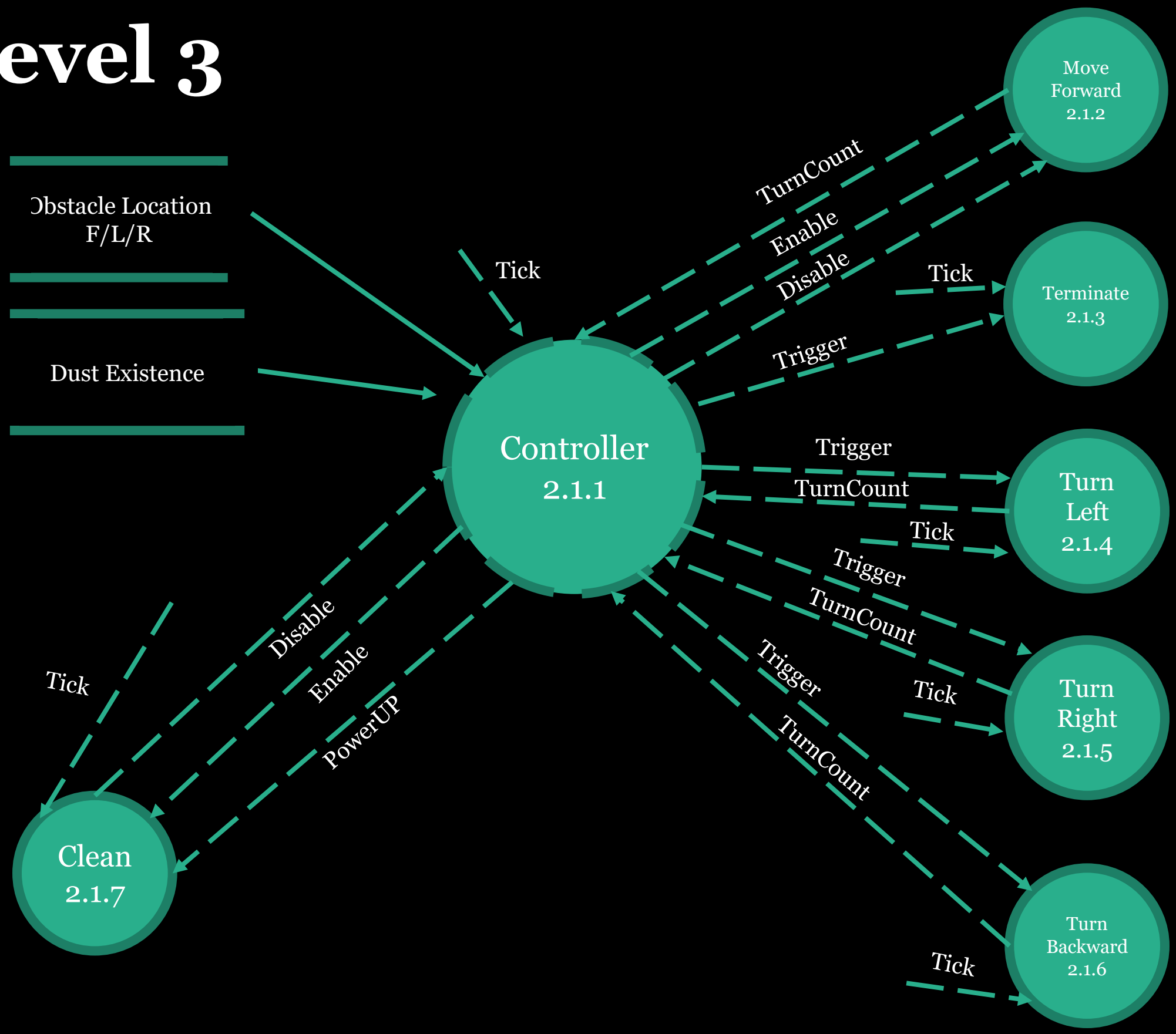
DFD Level 2 -
Process Specification

Reference No.	2.3
Name	Cleaner Interface
Input	Cleaner Command
Output	Clean , PowerUp
Process Description	Cleaner Command를 받아, Clean과 PowerUP에 할당

DFD Level 2 - Data Dictionary

Input / Output Event	Description	Format / Type
Front Sensor Input	Detects obstacles in front of the RVC	True/False, Interrupt
Left Sensor Input	Detects obstacles in the left side of the RVC periodically	True/False, Periodic
Right Sensor Input	Detects obstacles in the right side of the RVC periodically	True/False, Periodic
Dust Sensor Input	Detects dust on the floor periodically	True/False, Periodic
Front Obstacle	Obstacles in front of the RVC	True/False, Periodic
Left Obstacle	Obstacle on the left side of the RVC	True/False, Periodic
Right Obstacle	Obstacle on the right side of the RVC	True/False, Periodic
Dust Existence(1)	Dust detected in front of RVC by Dust Sensor	True/False, Periodic
Obstacle Location	Save Obstacle Location Data	Forward / Left / Right / Stop / Backward
Motor Command	Direction Commands	Forward / Left / Right / Stop / Backward
Dust Existence(2)	Save Dust Existence Data	True/False
Cleaner Command	Cleaner Turn off / Turn on	On / Off
Direction	Direction commands to the motor (go forward / turn left with an angle / turn right with an angle)	Forward / Left / Right / Stop / Backward
Terminate	RVC terminated when TurnCount rises above a certain level	trigger
Clean	Turn off / Turn on	On / Off
PowerUp	PowerUp for stronger cleaning	On / Off
TurnCount	Record the number of times the RVC Rotate to prevent an infinite Loop if the RVC goes forward, initialize to zero	integer

Level 3


Obstacle Location
F/L/R
Dust Existence
Tick
Controller
2.1.1
TurnCount
Enable
Disable
Trigger
Move
Forward
2.1.2
Tick
Terminate
2.1.3
Trigger
TurnCount
Turn
Left
2.1.4
Tick
Trigger
TurnCount
Turn
Right
2.1.5
Tick
Trigger
TurnCount
Turn
Backward
2.1.6
Tick
Tick
Disable
Enable
PowerUP
Clean
2.1.7

DFD Level 3 -
Process Specification

Reference No.	2.1.1
Name	Controller
Input	Obstacle Location, Dust Existence, Tick, TurnCount
Output	Trigger, Enable, Disable, PowerUP, Tick
Process Description	Obstacle Location, Dust Existence, Tick, TurnCount를 받아 RCV의 동작을 결정. TurnCount는 최초 작동시 0으로 초기화되며, Controller가 입력을 받을 때마다 1씩 증가한다. Move Forward로부터 입력을 받을 경우 0으로 초기화된다.

Reference No.	2.1.2
Name	Move Forward
Input	Enable , Disable
Output	TurnCount, Motor Command
Process Description	Controller에게 받은 입력을 통해, Motor Interface에 동작 전달 Command에게 “TurnCount를 0으로 초기화”를 전달

DFD Level 3 -
Process Specification

Reference No.	2.1.3
Name	Terminate
Input	Trigger, Tick
Output	Motor Command
Process Description	TurnCount가 100 이상이면, RVC의 전원을 종료

Reference No.	2.1.4
Name	Turn Left
Input	Trigger, Tick
Output	TurnCount, Motor Command
Process Description	Controller에게 받은 입력을 통해, Motor Interface에 동작을 전달 Command에게 “TurnCount에 1을 더함”을 전달

DFD Level 3 -
Process Specification

Reference No.	2.1.5
Name	Turn Right
Input	Trigger, Tick
Output	TurnCount, Motor Command
Process Description	Controller에게 받은 입력을 통해, Motor Interface에 동작을 전달 Command에게 “TurnCount에 1을 더함”을 전달

Reference No.	2.1.6
Name	Turn Backward
Input	Trigger, Tick
Output	TurnCount, Motor Command
Process Description	Controller에게 받은 입력을 통해, Motor Interface에 동작을 전달 Command에게 “TurnCount에 1을 더함”을 전달

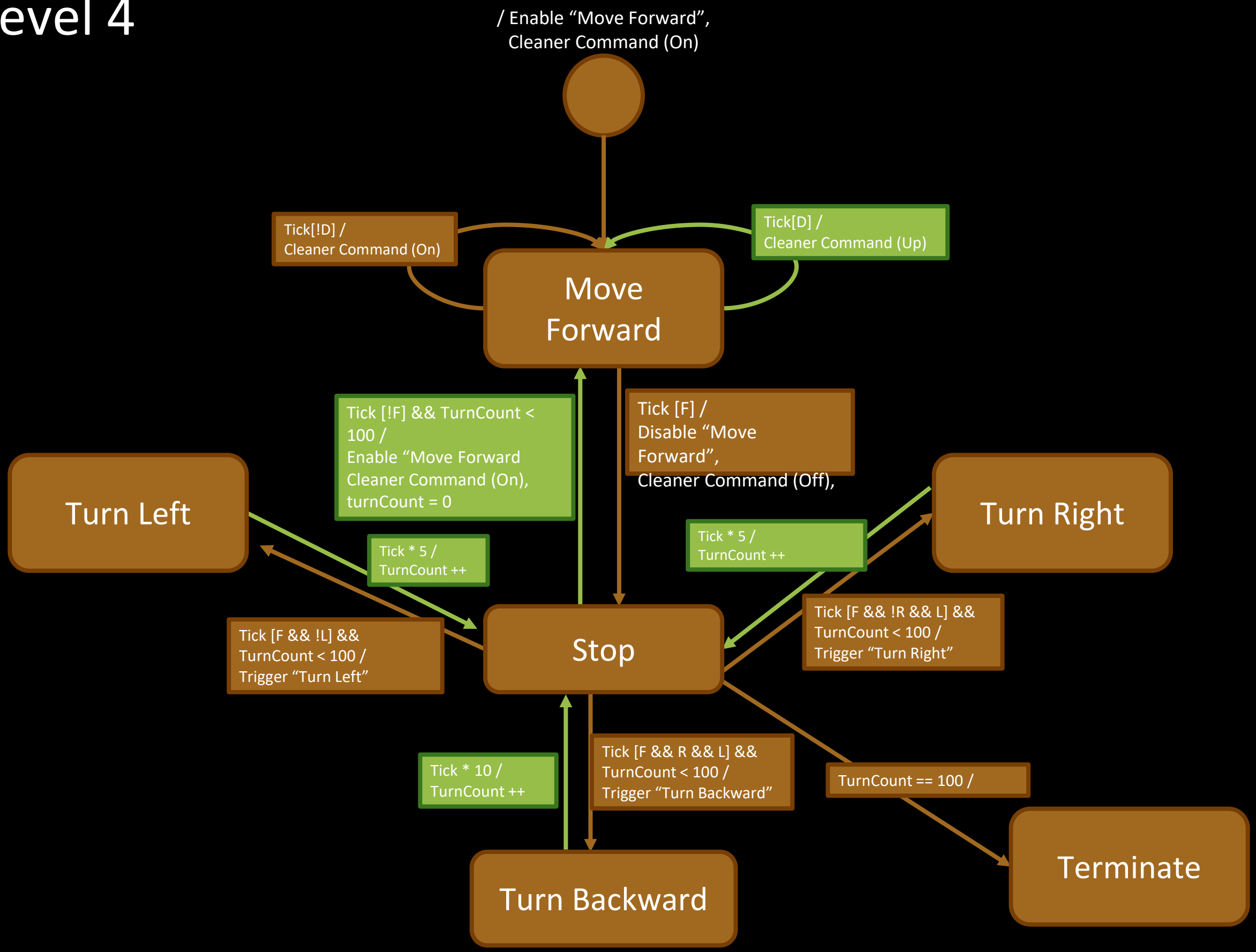
DFD Level 3 -
Process Specification

Reference No.	2.1.7
Name	Clean
Input	Disable , Enable , PowerUp , Tick
Output	Clean Command
Process Description	Controller에게 받은 입력을 통해, Clean Interface에 동작을 전달 PowerUp 시, Cleaner의 출력을 높인다.

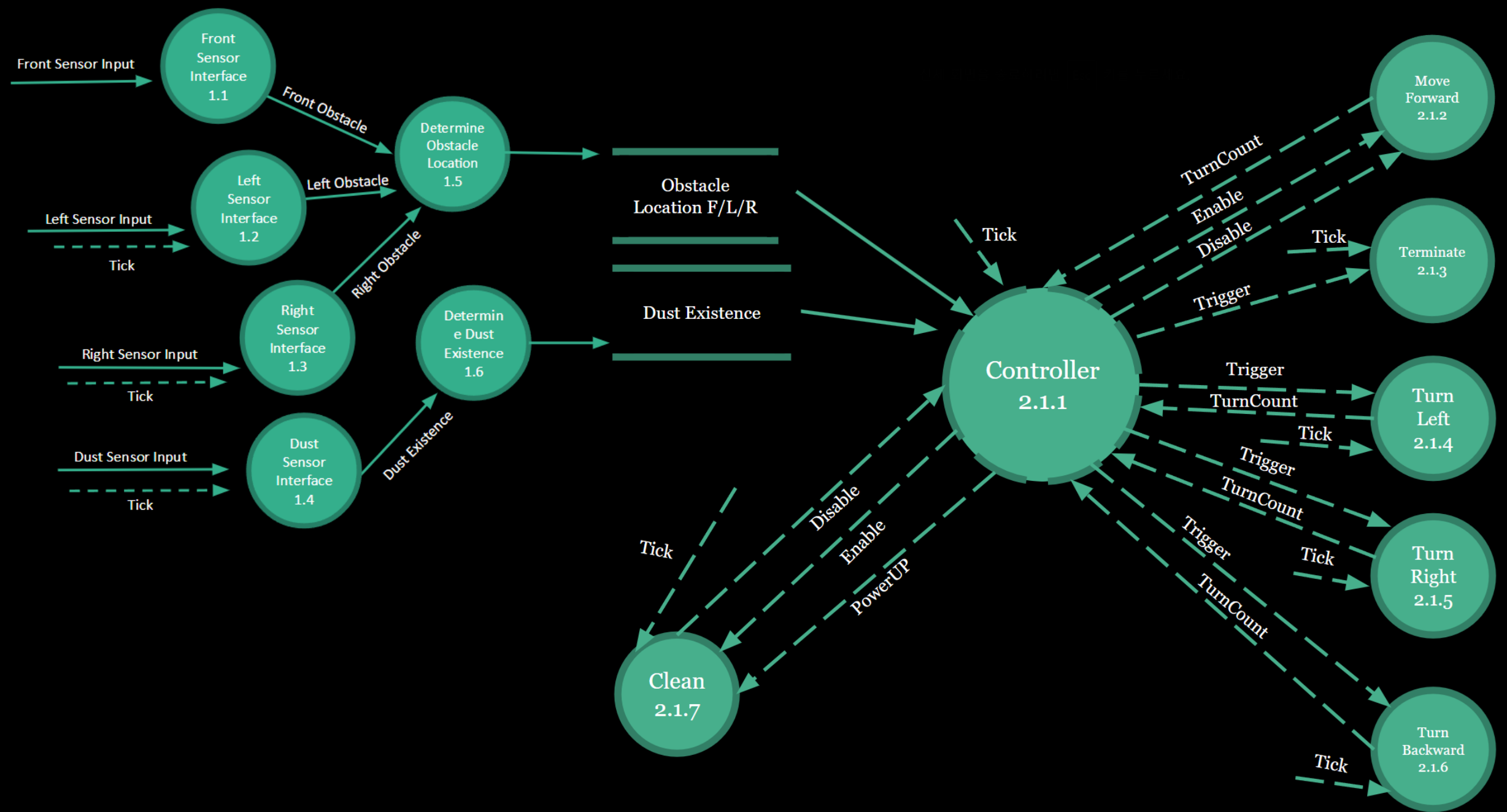
DFD Level 3 - Data Dictionary

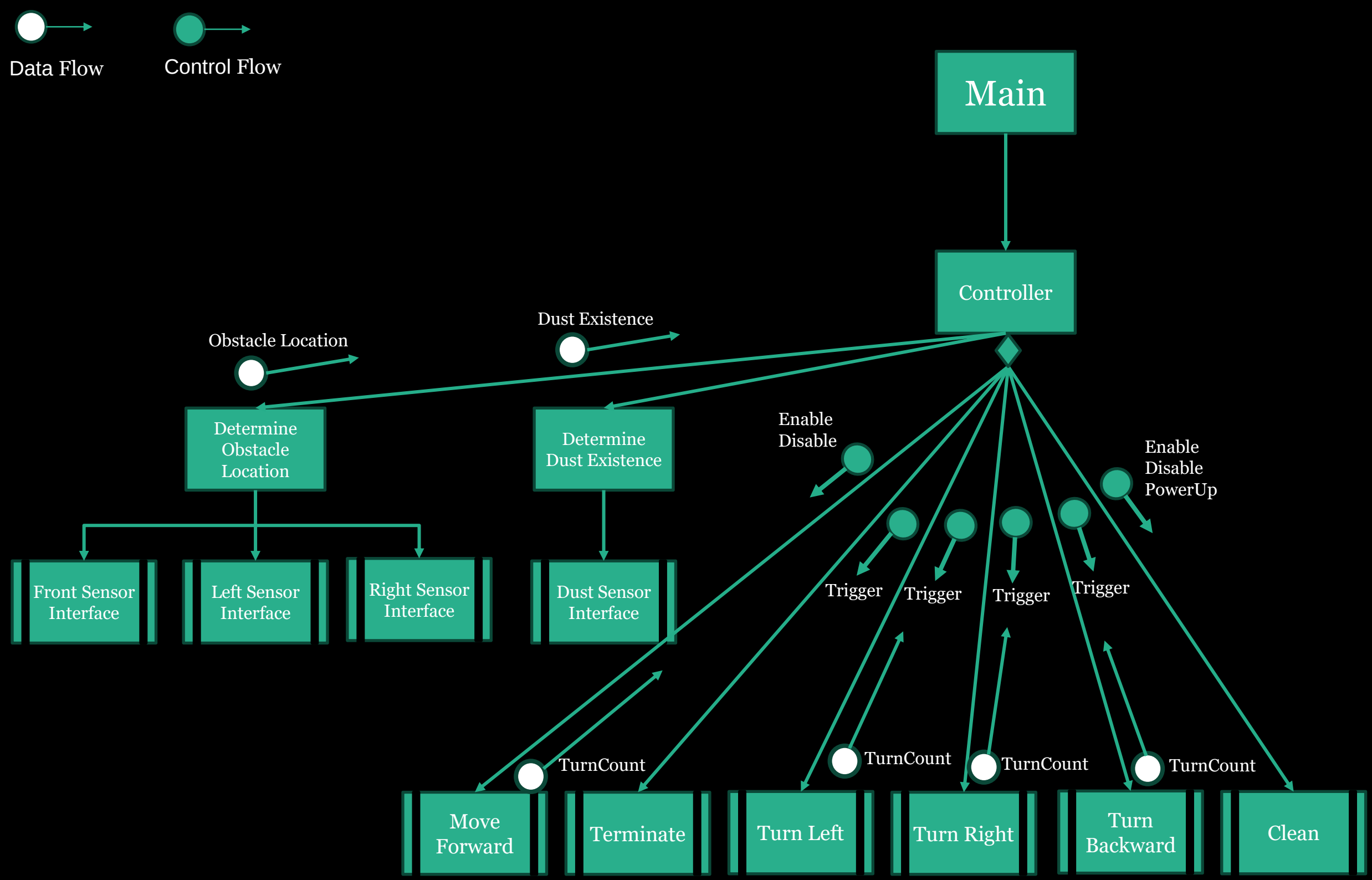
Input / Output Event	Description	Format / Type
Front Sensor Input	Detects obstacles in front of the RVC	True/False, Interrupt
Left Sensor Input	Detects obstacles in the left side of the RVC periodically	True/False, Periodic
Right Sensor Input	Detects obstacles in the right side of the RVC periodically	True/False, Periodic
Dust Sensor Input	Detects dust on the floor periodically	True/False, Periodic
Front Obstacle	Obstacles in front of the RVC	True/False, Periodic
Left Obstacle	Obstacle on the left side of the RVC	True/False, Periodic
Right Obstacle	Obstacle on the right side of the RVC	True/False, Periodic
Dust Existence(1)	Dust detected in front of RVC by Dust Sensor	True/False, Periodic
Obstacle Location F/L/R	Save Obstacle Location Data(Forward / Left / Right)	True/False
Motor Command	Direction Commands	Forward / Left / Right / Stop / Backward
Dust Existence(2)	Save Dust Existence Data	True/False
Cleaner Command	Cleaner Turn off / Turn on	On / Off
Direction	Direction commands to the motor (go forward / turn left with an angle / turn right with an angle)	Forward / Left / Right / Stop / Backward
Terminate	RVC terminated when TurnCount rises above a certain level	trigger
Clean	Turn off / Turn on	On / Off
PowerUp	PowerUp for stronger cleaning	On / Off
TurnCount	Record the number of times the RVC Rotate to prevent an infinite Loop if the RVC goes forward, initialize to zero	integer

Level 4


/ Enable “Move Forward”,
Cleaner Command (On)
Tick[!D] /
Cleaner Command (On)
Tick[D] /
Cleaner Command (Up)
Move Forward
Tick [!F] && TurnCount < 100 /
Enable “Move Forward
Cleaner Command (On),
turnCount = 0
Tick [F] /
Disable “Move Forward”,
Cleaner Command (Off),
Turn Left
Turn Right
Tick * 5 /
TurnCount ++
Tick * 5 /
TurnCount ++
Tick [F && !L] &&
TurnCount < 100 /
Trigger “Turn Left”
Stop
Tick [F && !R && L] &&
TurnCount < 100 /
Trigger “Turn Right”
Tick * 10 /
TurnCount ++
Tick [F && R && L] &&
TurnCount < 100 /
Trigger “Turn Backward”
TurnCount == 100 /
Turn Backward
Terminate

DFD - Overview




Data Flow
Control Flow
Main
Controller
Obstacle Location
Dust Existence
Determine Obstacle Location
Determine Dust Existence
Enable Disable
Enable Disable PowerUp
Front Sensor Interface
Left Sensor Interface
Right Sensor Interface
Dust Sensor Interface
Trigger
Trigger
Trigger
Trigger
TurnCount
TurnCount
TurnCount
TurnCount
Move Forward
Terminate
Turn Left
Turn Right
Turn Backward
Clean

Specification Analysis

- 전방에 장애물이 없으면 Cleaner Mode를 On으로 전환하고 직진한다.
  직진하는 동안 Dust Sensor가 먼지를 감지하면,
  Cleaner Mode를 Power up 모드로 전환하여 출력을 강화한다.
  전방에 Dust가 더이상 감지되지 않으면 Power up 모드를 On으로 전환한다

- 전방에 장애물이 감지되어 기기가 직진을 할 수 없을 때에는 기기를 정지하고,
  Cleaner Mode를 Off로 전환하여 청소 동작을 일시적으로 정지시킨다.
  기기의 왼쪽과 오른쪽에 장애물이 있는지 센서를 Periodic하게 작동시켜 탐지한다.

- 왼쪽에 장애물이 감지되지 않는다면 기기를 제자리에서 왼쪽으로 90도 회전하며,
  오른쪽에 장애물이 감지되지 않는다면 기기를 제자리에서 오른쪽으로 90도 회전한다.

- 양쪽 모두에 장애물이 감지되지 않는다면 왼쪽으로 회전한다.

- 왼쪽과 오른쪽 모두 장애물이 감지되면, 기기를 제자리에서 왼쪽으로 180도 회전한다.

- 직진 동작 없이 회전 동작이 100회 이상 연속으로 발생할 경우,
  기기가 정상 작동하지 못하는 상황으로 판단하여 기기의 전원을 종료한다.